

Machine Learning Final Project

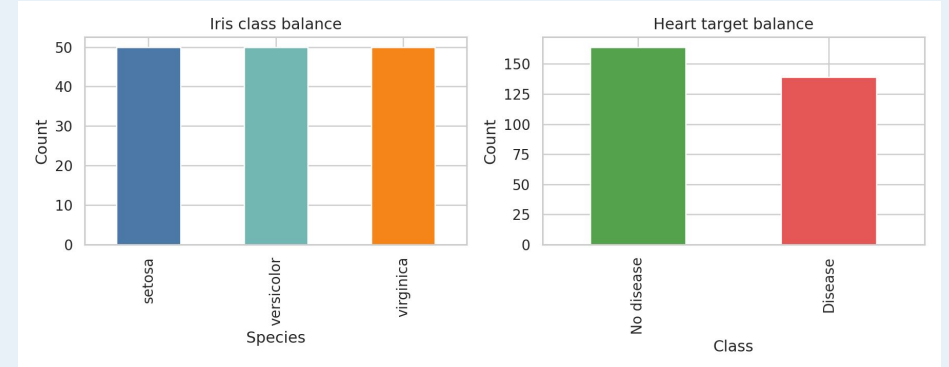
Comparing Logistic Regression, KNN, and Decision Trees on Iris and Heart Disease Data

Datasets

Iris
Heart Disease

Models

Logistic Regression
KNN
Decision Tree



What the project had to do

1

1. Explore data

Describe features, target, data types, and why the datasets were chosen.

2

2. Preprocess

Handle encoding, scaling, train/test split, and missing values.

3

3. Train models

Run 2–3 supervised algorithms for each dataset using scikit-learn.

4

4. Tune + compare

Use GridSearchCV and compare metrics across models and datasets.

5

5. Evaluate + relate to literature

Interpret metrics, plots, strengths, limits, and compare with prior work.

Datasets at a glance

Iris

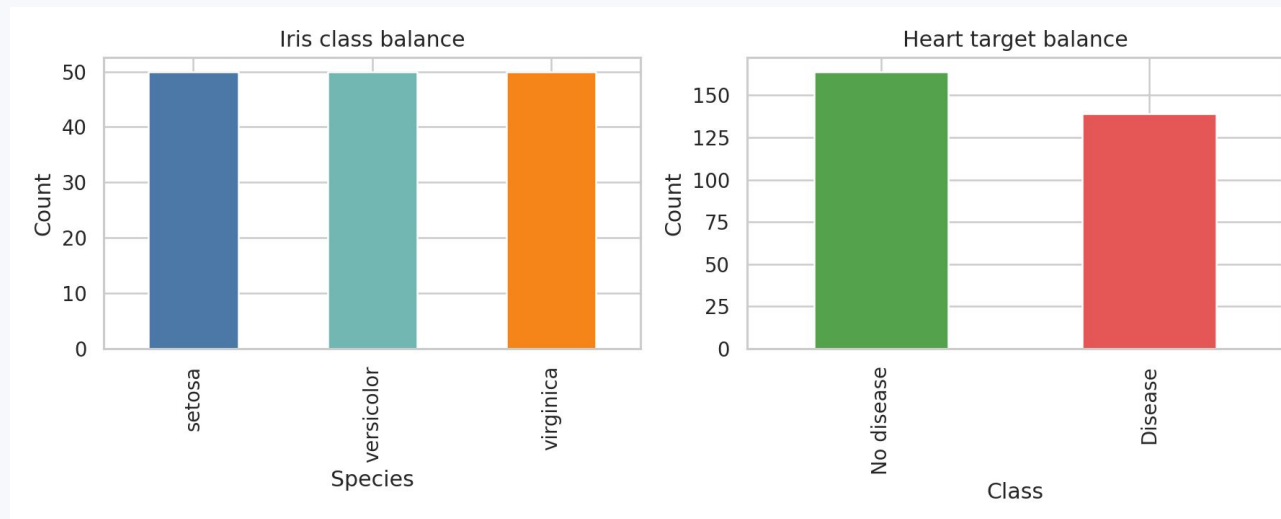
150 samples
4 numeric features
3 balanced classes
No missing values
Built into scikit-learn

Heart Disease

303 records
13 predictors + binary target
Mixed numeric/categorical fields
Small amount of missing data in Ca/Thal

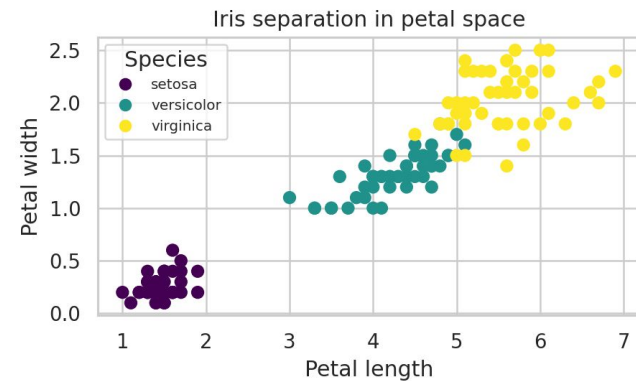
Why these datasets?

Iris gives a clean benchmark for multiclass classification.
Heart Disease adds a realistic healthcare problem with more feature complexity.



Takeaway: both datasets are small enough to explain clearly, but different enough to show how model behavior changes with task complexity.

Dataset 1: Iris exploration



Iris feature correlations

sepal length (cm)	1.00	-0.12	0.87	0.82
sepal width (cm)	-0.12	1.00	-0.43	-0.37
petal length (cm)	0.87	-0.43	1.00	0.96
petal width (cm)	0.82	-0.37	0.96	1.00

sepal length (cm)

sepal width (cm)

petal length (cm)

petal width (cm)

What we see

Petal features separate classes much better than sepal features. Correlation is strongest between petal length and petal width.

Preprocessing

No missing data. StandardScaler used for Logistic Regression and KNN; not needed for Decision Tree.

Why keep it?

It is a classic baseline dataset, so model differences are easy to interpret and explain.

Dataset 2: Heart Disease exploration

What we see

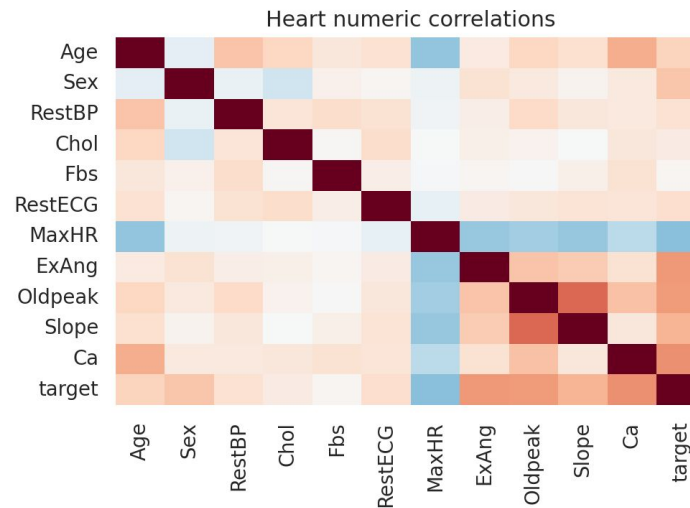
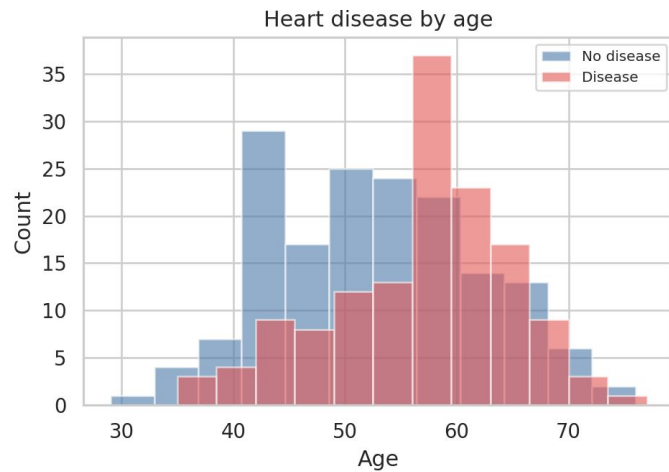
Patients with disease skew older on this sample. No single numeric feature is enough by itself, so the task is harder.

Preprocessing

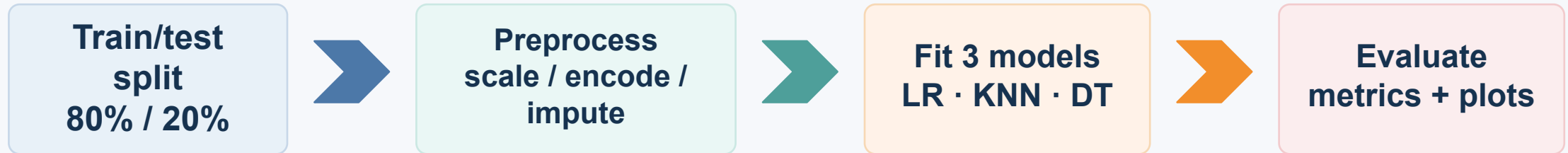
Binary target from AHD. One-hot encoding for ChestPain and Thal, median/mode imputation, scaling for LR and KNN.

Why keep it?

It is clinically meaningful and better reflects messy real-world tabular data.



Modeling workflow



Why each model?

Logistic Regression: interpretable linear baseline

KNN: distance-based model that often works well on small numeric datasets

Decision Tree: nonlinear and easy to explain

Validation

Stratified train/test split for both datasets

GridSearchCV for hyperparameter tuning

Metrics: accuracy, precision, recall, F1, confusion matrix, ROC-AUC (heart)

What changed by dataset?

Iris used only scaling.

Heart Disease needed encoding and imputation before fitting non-tree models.

Hyperparameter tuning summary

Dataset	Model	Best parameters	CV accuracy
Iris	Logistic Regression	{'model__C': 10}	0.967
Iris	KNN	{'model__n_neighbors': 3, 'model__weights': 'uniform'}	0.967
Iris	Decision Tree	{'max_depth': 3, 'min_samples_split': 2}	0.958
Heart	Logistic Regression	{'model__C': 10}	0.843
Heart	KNN	{'model__n_neighbors': 3, 'model__weights': 'uniform'}	0.810
Heart	Decision Tree	{'max_depth': 3, 'model__min_samples_split': 2}	0.773

What improved?

Best cross-validation scores stayed very high on Iris ($\approx 0.96+$). On Heart Disease, Logistic Regression tuned best by CV, but KNN tied it on the held-out test set.

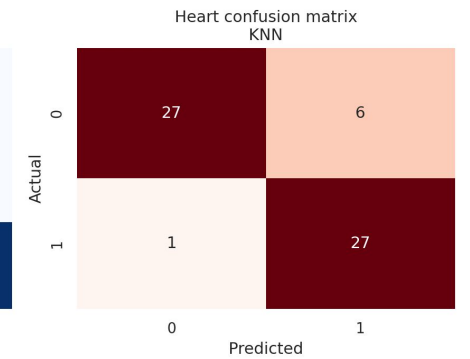
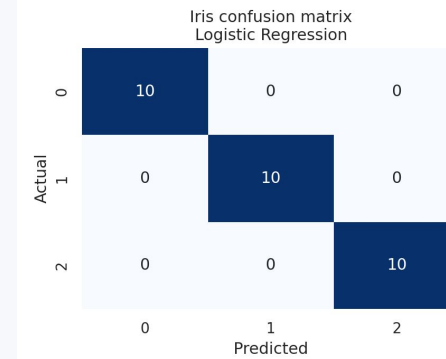
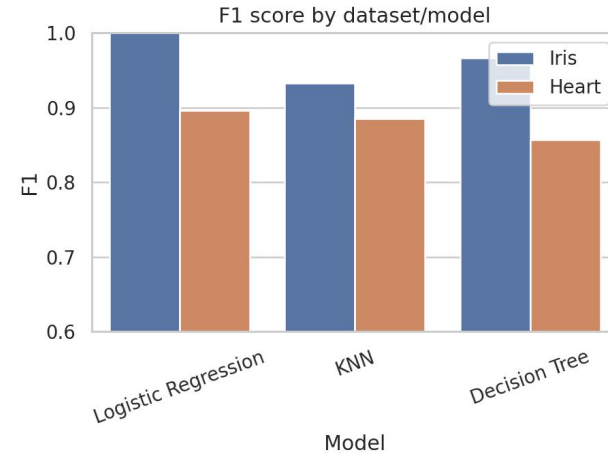
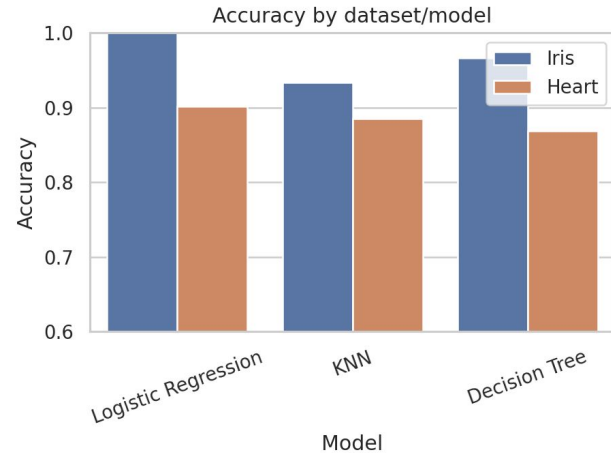
Why these settings?

Smaller K values preserved local structure on both datasets. Shallow trees controlled overfitting. Larger C made Logistic Regression less regularized.

Main lesson

Tuning mattered more on the harder heart dataset than on Iris, where all three models were already near ceiling.

Results: Iris dataset



Performance

All three models reached 93.3% test accuracy; the dataset is simple enough that model choice mattered less than clean features.

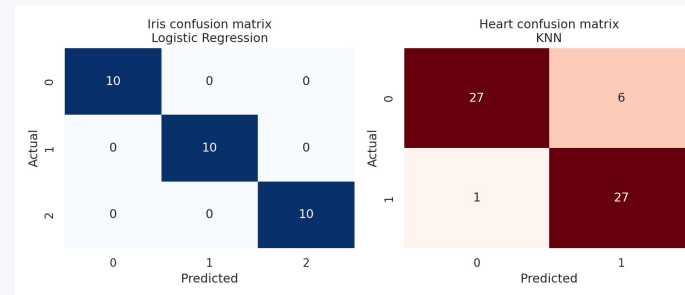
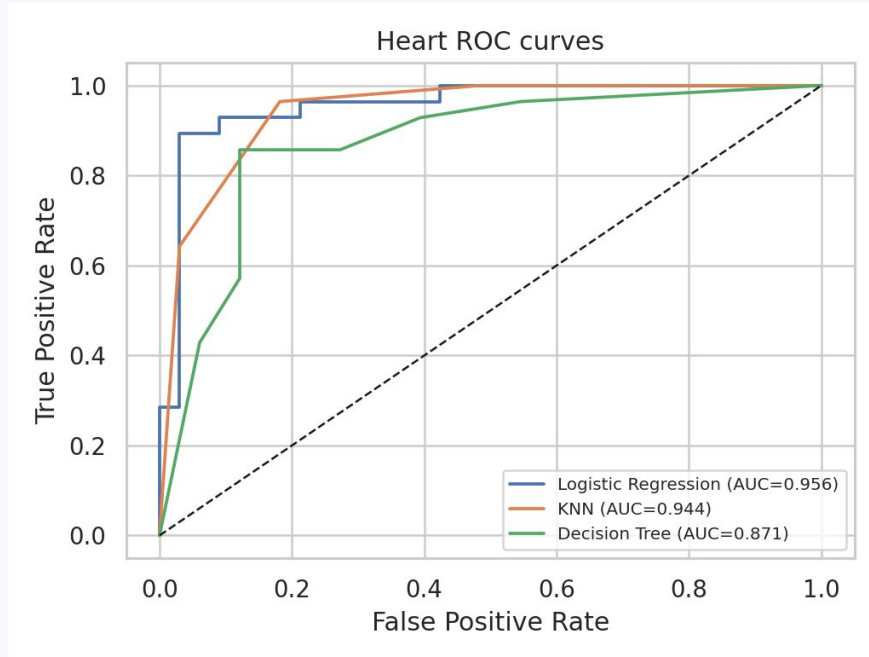
Best model to present

Logistic Regression is the easiest to justify here: same accuracy as others, cleaner interpretation, and a perfect-looking confusion matrix.

Interpretation

This dataset is close to linearly separable in petal space, so even simple models perform extremely well.

Results: Heart Disease dataset



Best on test set

KNN and Logistic Regression both reached 90.2% accuracy.

Best ROC-AUC

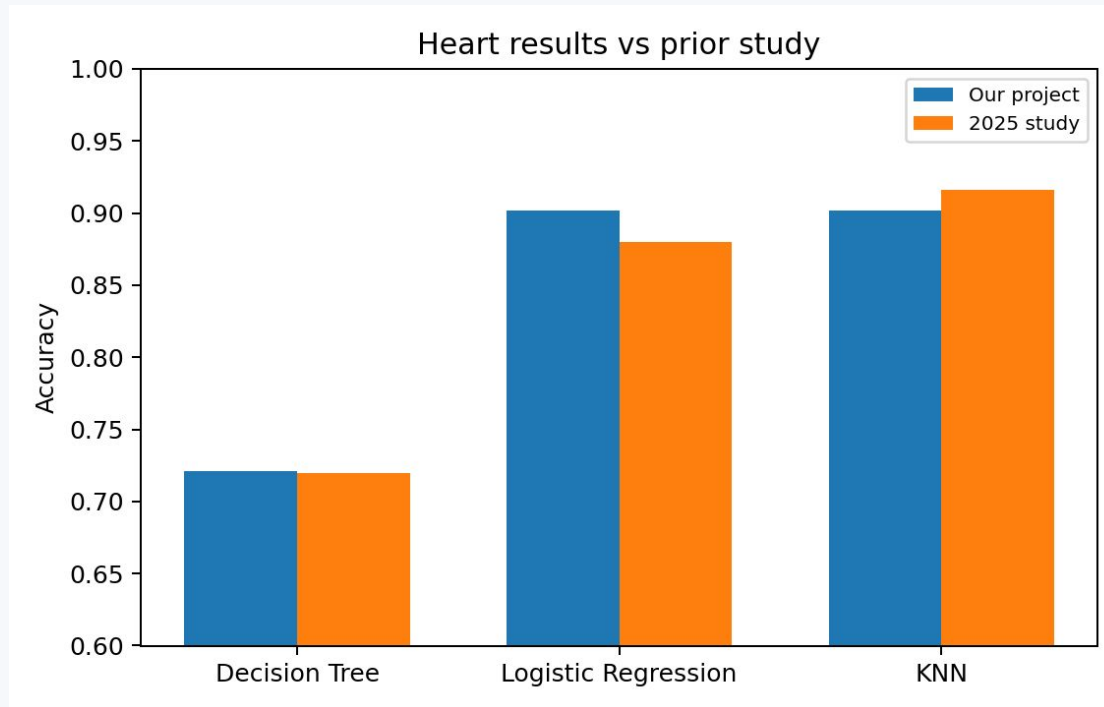
KNN: 0.965
Logistic Regression: 0.957

Weakest model

Decision Tree dropped to about 72% accuracy and much lower AUC.

Why the drop? The heart dataset is smaller, noisier, and more mixed in feature type, so simple splits lose information and overfit faster.

Comparison to prior published work



Prior work used

Rachmat et al. (2025) compared Decision Tree, KNN, Logistic Regression, Random Forest, XGBoost, and Stacking on the UCI Heart Disease dataset.

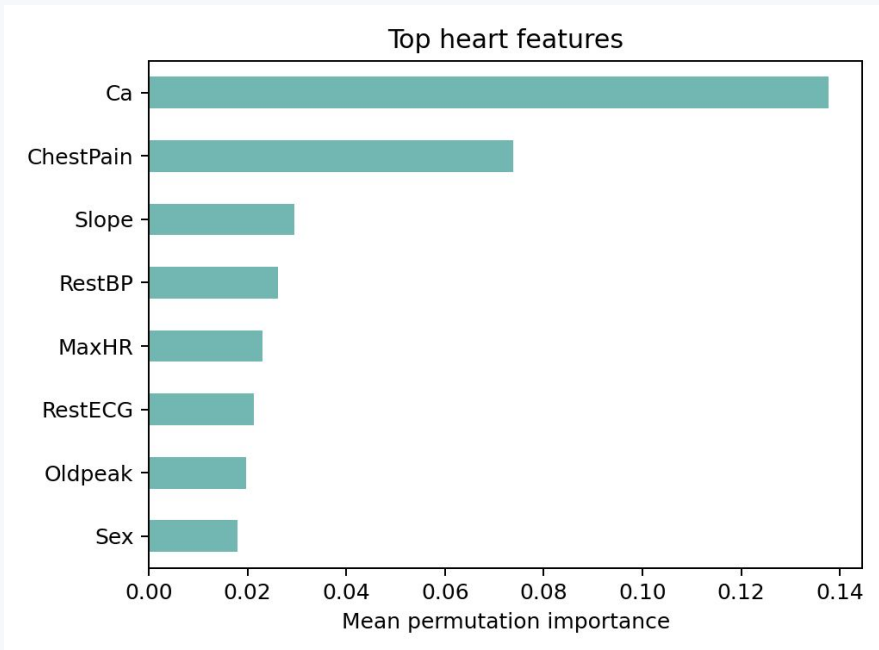
Reported results

They reported Logistic Regression accuracy around 0.88, Decision Tree around 0.72, and stronger AUC from stacking/logistic approaches.

Our comparison

Our Logistic Regression was slightly higher on this split (0.902), KNN was a bit lower than their KNN (0.916 vs 0.902), and Decision Tree was essentially the same.

Interpretation, limits, and real-world relevance



What the model is learning

For Heart Disease, vessel count (Ca), chest pain type, slope, resting blood pressure, and max heart rate were among the strongest signals in our run.

Strengths

- Classical models are fast, transparent, and easy to justify.
- Results are strong on small datasets.
- Workflow is reproducible in scikit-learn.

Limitations

- Small samples limit generalization.
- One random split can change results.
- More feature engineering or ensembles could improve heart predictions.

Real-world relevance: interpretable tabular models can support early screening, but they should assist clinicians rather than replace expert judgment.

Conclusion

1. We met the project brief

2 datasets, 3 supervised models each, preprocessing, GridSearchCV, evaluation metrics, plots, and literature comparison.

2. Main result

Iris was solved well by every model. Heart Disease separated stronger and weaker models more clearly.

3. Best practical choice

Logistic Regression gave the best balance of performance and interpretability across the whole project.

Final takeaway

Classical supervised learning can still perform very well on structured tabular data when the workflow is careful: understand the data, preprocess correctly, tune key parameters, and evaluate with the right metrics.

Thank you